Sham Satish Thakare

Independent Computer Science Researcher & ML Software Engineer

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RESEARCH STATEMENT & INTERESTS

I conduct systems and empirical research on **Reliable Adaptive Intelligent Systems**. My work focuses on four principal vectors:

1. **Calibrated Reasoning & Adaptive Computation**: Diagnostic probes, token entropy credit assignment, and RLVR self-consistency calibration.
2. **Long-Horizon Agent Reliability & Tool Safety**: State failure recovery protocols, temporal post-recovery action divergence, and policy enforcement.
3. **Learning-Augmented Fault-Tolerant Systems**: Raft dynamic quorum adaptation (AdaptiveReplica) with uncertainty-aware trust gates.
4. **Verifiable, Private & Observable AI Systems**: Zero-knowledge provenance graph proofs (EnclaveShield) and graph-constrained causal walks (TraceMind).

EDUCATION

Anantrao Pawar College of Engineering & Research (APCOER), Pune, India
Bachelor of Technology in Artificial Intelligence and Data Science **Jan 2020 – Jun 2024**

- **Cumulative GPA**: 8.70 / 10.00
- **Academic Honors**: Awarded 100% Merit Scholarship for 8 consecutive semesters based on academic performance.
- **Core Coursework**: Algorithms, Operating Systems, Data Structures, Computer Networks, Linear Algebra, Probability & Statistics, Database Management Systems, Machine Learning, Computational Modeling.

SUBMITTED MANUSCRIPTS & PREPRINTS

- **When Confidence Proxies Confound Reasoning Complexity: Pitfalls of Uncertainty-Weighted Credit Assignment in LM RL**
  Sham Satish Thakare **Submitted to IEEE TAI (Aug 2026)**
  Exposes length confounding in token predictive entropy ($r = +0.486$), proving sample-level uncertainty weighting (CA-GRPO) yields zero Pass@1 gain over standard outcome GRPO. Code: `https://github.com/shamddd/ear_grpo_reasoning`
- **recovery-eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in LM Reasoning**
  Sham Satish Thakare **Submitted to IEEE BigData 2026 (ID: BigD497)**
  Introduces a matched state-contrast protocol ($D_{\text{recovery}} = -0.1100$, 95% CI [-0.240, +0.030]), demonstrating Instruct checkpoints lack a recovery-specific advantage over Base checkpoints. Code: `https://github.com/shamddd/recovery_eval`
- **Amortized Intervention Frontiers for Language-Model Reasoning: When Does Training Beat Search?**
  Sham Satish Thakare **Submitted to TMLR / NeurIPS Workshop (Aug 2026)**
  Formalizes deployment cost frontiers ($C_{\text{total}} = C_{\text{train}} + Q \cdot C_{\text{inference}}$). Demonstrates OOD length extrapolation accelerates RLVR amortization crossover ($R_f \approx 0.0618 \ll 1.0$). Code: `https://github.com/shamddd/submission_package`
- **StateShift: State-Matched Evaluation of Error Recovery in Neural Reasoning**
  Sham Satish Thakare **Submitted to Elsevier Artificial Intelligence (AIJ 2026)**
  Formulates verifier-defined continuation state matching on neural reasoning rollouts. Code: `https://github.com/shamddd/stateshift`
- **AdaptiveReplica: Dynamic Quorum Adaptation and Failure-Aware Replica Selection in Distributed Consensus**

Sham Satish Thakare **Preprint / Working Paper (Target: IEEE TPDS)**
Engineers Raft joint-consensus dynamic vote weighting with trust gates, reducing p99 write latency to 13.50ms (88.8% reduction vs static 120.48ms) with zero stale reads. Code: https://github.com/shamddd/quorumshift

- **EnclaveShield: Zero-Knowledge Memory Attestation and Side-Channel Mitigation for Hardware Enclaves**
  Sham Satish Thakare **Preprint / Working Paper (Target: IEEE TDSC)**
  ZK remote attestation proofs and adaptive Path ORAM rebalancing, bounding access latency to 1.47ms ($H(A) = 0.82 \pm 0.02$). Code: https://github.com/shamddd/enclaveshield

PROFESSIONAL EXPERIENCE

Spectra Corporate Service, Pune, India
Data Engineer / Machine Learning Specialist **Jun 2024 – Present**

- Architected scalable data pipelines using C++, Python, and SQL on GCP and AWS (GKE, Compute Engine, Cloud Storage).
- Implemented Linux multithreaded system software, synchronization primitives, container security controls, and IaC pipelines using Terraform, Docker, Kubernetes, and GitHub Actions.
- Constructed Prometheus and Grafana distributed telemetry dashboards for SLA monitoring and proactive incident detection.

APCOER Pune, Pune, India
Research Intern / Teaching Fellow **Jun 2023 – Dec 2024**

- Served as Teaching Fellow for *Intro to Algorithms and Their Limitations* and *Foundations of ML: AI Alignment and Safety* (Aug–Dec 2024).
- Served as Course Producer for *Fundamentals of Computation*, *Introduction to Algorithms*, and *Theory of Computation* (Aug 2022–May 2023).
- Assisted faculty in experimental design, statistical analysis, and open-source reproducibility protocols.

TECHNICAL SKILLS

- **Languages**: Python, C++, C, Rust, Go, Java, SQL, Bash, LaTeX
- **AI & Machine Learning**: PyTorch, GRPO/PPO RL Probes, Transformers, HuggingFace, NumPy, SciPy, Scikit-Learn
- **Systems & Security**: Raft Consensus, OpenTelemetry, TEEs (SGX/TDX), Path ORAM, Zero-Knowledge Proofs
- **Cloud & Infrastructure**: Docker, Kubernetes, Helm, Terraform, GCP, AWS, Prometheus, Grafana, Linux Kernel/POSIX